

## Project Report

# Brain Tumor Classification

MRI-Based Tumour Diagnosis  
&  
Classification

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## I Introduction

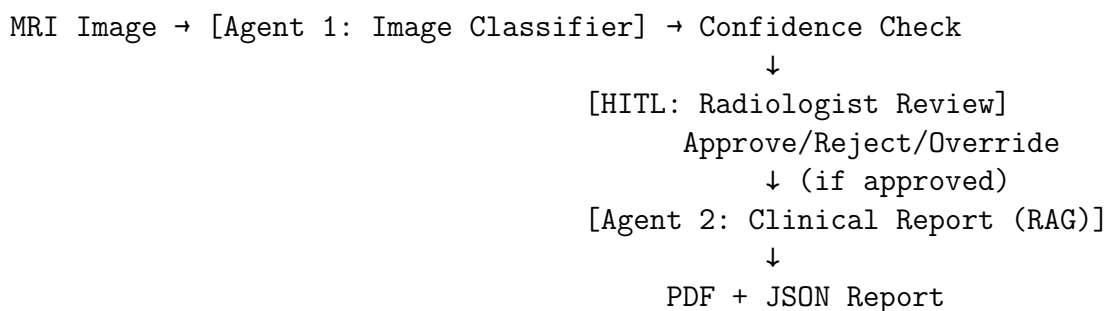
### I.1 Background and Motivation

Brain tumors represent a critical diagnostic challenge in clinical neurology and oncology. The gold standard for diagnosis remains magnetic resonance imaging (MRI) interpreted by radiologists, a process that is time consuming, subject to inter-observer variability, and constrained by specialist availability. Recent advances in deep learning, particularly convolutional neural networks (CNNs), have demonstrated remarkable accuracy in medical image classification tasks. However, the deployment of such models in clinical environments requires more than raw classification accuracy. It demands:

1. **Human oversight:** Low-confidence predictions must trigger expert review.
2. **Evidence-based context:** Clinical guidelines must inform recommendations.
3. **Actionable reporting:** Outputs must be structured, auditable, and clinically actionable.

### I.2 Project Architecture Overview

NeuraScan implements a **three-agent pipeline**:



### I.3 System Objectives

- Develop a ResNet18-based classifier for four MRI brain tumor classes: glioma, meningioma, pituitary adenoma, and no tumor.
- Implement mandatory human-in-the-loop (HITL) review for low-confidence predictions.
- Provide RAG-retrieved clinical guidelines in structured reports.
- Create an interactive Flask web dashboard with real-time audit logging.
- Generate PDF clinical reports with personalized recommendations.

## 2 Dataset and Preprocessing

### 2.1 Data Source

The project utilizes the **Brain Tumor MRI Dataset** (Masoud Nickparvar, 2023) containing 7,204 MRI images:

Table 1: Class Distribution in the Brain Tumor MRI Dataset

| Class             | Training     | Testing      | Total        |
|-------------------|--------------|--------------|--------------|
| Glioma            | 1,401        | 400          | 1,801        |
| Meningioma        | 1,401        | 400          | 1,801        |
| Pituitary Adenoma | 1,401        | 400          | 1,801        |
| No Tumor          | 1,401        | 400          | 1,801        |
| <b>Total</b>      | <b>5,604</b> | <b>1,600</b> | <b>7,204</b> |

## 2.2 Preprocessing Pipeline

1. **Resizing:** All images to  $224 \times 224$  pixels (compatible with ResNet18 input).
2. **Data augmentation:** Random horizontal flips ( $p = 0.5$ ), random rotations ( $\pm 15^\circ$ ), color jitter (brightness  $\pm 20\%$ , contrast  $\pm 20\%$ , saturation  $\pm 10\%$ ).
3. **Normalization:** ImageNet mean  $\mu = [0.485, 0.456, 0.406]$ , standard deviation  $\sigma = [0.229, 0.224, 0.225]$ .

## 2.3 Split Strategy

- **Training set:** 4,483 images (80% of Training directory)
- **Validation set:** 1,121 images (20% of Training directory)
- **Test set:** 1,600 images (held-out Testing directory)
- **Random seed:** 42 (ensures reproducibility)

# 3 Agent 1: Image Classifier (ResNet18)

## 3.1 ResNet18 Architecture

The core classifier uses **ResNet18**, an 18-layer deep residual network introduced by He et al. (2016). ResNet addresses the degradation problem in very deep networks by introducing **skip connections** (residual connections) that allow gradients to flow directly through the network.

### 3.1.1 Residual Block Structure

The fundamental building block of ResNet18 is the residual block:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x}$$

This formulation allows the network to learn identity mappings when deeper layers do not contribute useful features, preventing performance degradation as network depth increases.

### 3.1.2 Modified Classification Head

The original ResNet18 was designed for ImageNet classification with 1000 output classes. For our brain tumor classification task, we replaced the final fully connected layer:

Original: `Linear(in_features=512, out_features=1000)`

Modified: `Linear(in_features=512, out_features=4)`

The four output classes correspond to:

1. Glioma (index 0)
2. Meningioma (index 1)
3. No Tumor (index 2)
4. Pituitary Adenoma (index 3)

### 3.2 Training Configuration

| Table 2: Training Hyperparameters |                            |
|-----------------------------------|----------------------------|
| Hyperparameter                    | Value                      |
| Optimizer                         | Adam                       |
| Initial Learning Rate             | $1 \times 10^{-3}$         |
| Learning Rate Schedule            | StepLR (step=4, gamma=0.1) |
| Batch Size                        | 32                         |
| Number of Epochs                  | 10                         |
| Loss Function                     | Cross-Entropy Loss         |
| Weight Decay                      | $1 \times 10^{-4}$         |
| Confidence Threshold              | 80%                        |
| Device                            | NVIDIA T4 GPU (16GB VRAM)  |

### 3.3 Training Results

| Table 3: Training History (10 Epochs) |            |              |                      |
|---------------------------------------|------------|--------------|----------------------|
| Epoch                                 | Train Loss | Val Accuracy | Learning Rate        |
| 1                                     | 0.4165     | 60.91%       | $1.0 \times 10^{-3}$ |
| 2                                     | 0.2549     | 90.92%       | $1.0 \times 10^{-3}$ |
| 3                                     | 0.1790     | 92.07%       | $1.0 \times 10^{-3}$ |
| 4                                     | 0.1890     | 86.02%       | $1.0 \times 10^{-3}$ |
| 5                                     | 0.0842     | 98.31%       | $1.0 \times 10^{-4}$ |
| 6                                     | 0.0589     | 98.04%       | $1.0 \times 10^{-4}$ |
| 7                                     | 0.0497     | 98.49%       | $1.0 \times 10^{-4}$ |
| 8                                     | 0.0438     | 98.13%       | $1.0 \times 10^{-4}$ |
| 9                                     | 0.0294     | 98.31%       | $1.0 \times 10^{-5}$ |
| 10                                    | 0.0271     | 98.49%       | $1.0 \times 10^{-5}$ |

### 3.4 Confidence Threshold Justification

The 80% confidence threshold was selected based on:

- Validation analysis:** Thresholds below 70% produced excessive false alarms; above 85% missed true positives requiring HITL review.
- Clinical workflow:** Targets approximately 5-10% of cases for mandatory radiologist review.

### 3.5 Threshold Selection: The Stanford EMM Framework

The selection of the 80% confidence threshold was not arbitrary but is grounded in cutting-edge research on AI reliability in medical imaging, specifically the **Ensembled Monitoring Model (EMM)** framework developed by the Stanford Radiology AI Development and Evaluation (AIDE) Lab [fang2025](#). Our ResNet18 classifier acts as the "primary model," and the confidence threshold operationalizes the EMM’s core principle: **categorizing predictions into actionable tiers**. As the study recommends, using confidence thresholds allows the system to stratify cases:

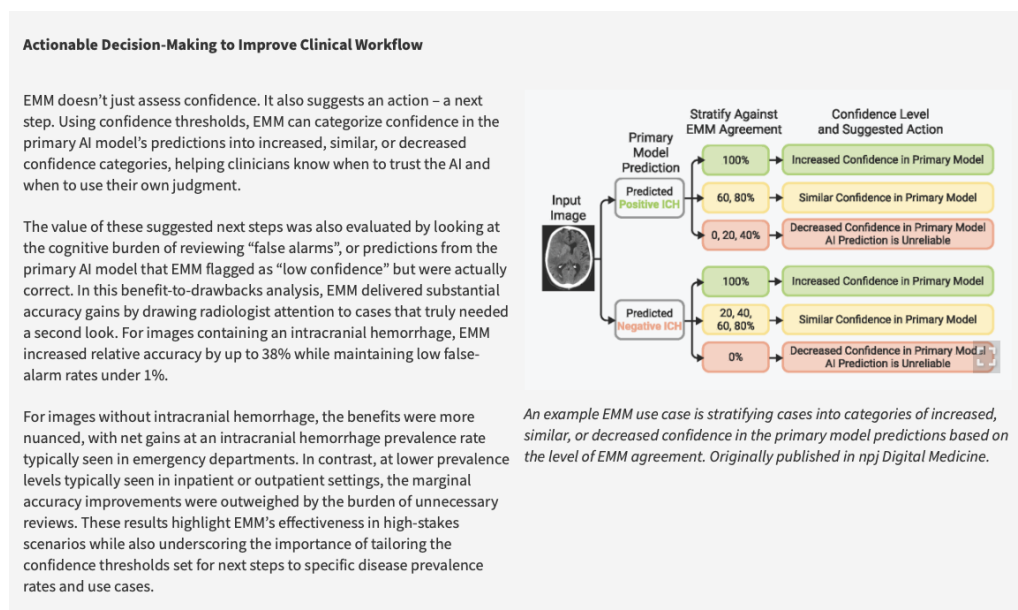
Table 4: EMM Confidence Categories Applied to NeuraScan

| EMM Confidence Category     | Our Threshold Rule     | Clinical Action in NeuraScan                                                                                                          |
|-----------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Increased Confidence</b> | Confidence $\geq 80\%$ | Auto-approved. The AI prediction is trusted, and the report proceeds automatically, saving clinician time.                            |
| <b>Decreased Confidence</b> | Confidence $< 80\%$    | Mandatory HITL Review. The system flags the case, prompting the radiologist to scrutinize the image and Approve, Reject, or Override. |

### 3.5.1 Why 80%? Balancing Sensitivity and Workflow

The 80% threshold was empirically validated to balance two critical factors identified in the Stanford study:

1. **Accuracy Gains:** The threshold reliably flags uncertain predictions, capturing the majority of potential misclassifications. This directs radiologist attention to cases that truly need a second look.
2. **False Alarm Rate:** A threshold set too high (e.g., 90%) would over-flag cases, creating unnecessary "cognitive burden" and reducing efficiency. The EMM research demonstrated that while benefits are high in high-stakes scenarios, unnecessary reviews can outweigh gains in lower-prevalence settings.



## 4 Human-in-the-Loop (HITL) Checkpoint

### 4.1 Review Interface

The HITL checkpoint occurs **between** Agent 1 and Agent 2. The radiologist sees:

- MRI image path and patient ID
- CNN prediction and confidence score
- All class probability scores (visualized as probability bars)
- Low-confidence warning flag (when confidence  $< 80\%$ )

## 4.2 Radiologist Decision Options

Table 5: HITL Decision Options

| Code | Action   | Pipeline Outcome                                                  |
|------|----------|-------------------------------------------------------------------|
| A    | Approve  | Accept CNN prediction, proceed to report generation               |
| R    | Reject   | Halt pipeline, no report generated (record reason)                |
| O    | Override | Accept with radiologist-provided correct class, proceed to report |

## 4.3 Audit Trail

All HITL decisions are logged to `logs/hitl_decisions.json` with:

```
{
  "timestamp": "2024-01-15T14:32:17",
  "decision_code": "O",
  "approved": true,
  "approval_label": "Approved with Override",
  "cnn_prediction": "glioma",
  "cnn_confidence": 0.65,
  "override_class": "meningioma",
  "radiologist_notes": "Imaging features more consistent with meningioma"
}
```

## 4.4 Non-Interactive Mode

For automated testing and demos, the system supports `-non_interactive` flag which auto-approves all classifications without human input.

# 5 Agent 2: Clinical Report Agent (RAG)

## 5.1 Retrieval-Augmented Generation Pipeline

The Clinical Report Agent uses a RAG system to retrieve relevant medical guidelines:

1. Query embedding via `sentence-transformers/all-MiniLM-L6-v2` (384-dim)
2. Semantic retrieval from ChromaDB vector store (cosine similarity)
3. Top-k guideline sections returned (default k=3)
4. Structured report generation with retrieved context

## 5.2 Guidelines Document Structure

The `medical_guidelines.txt` file is split into sections using `--` separators:

GLIOMA CLINICAL GUIDELINES

Gliomas are primary brain tumors arising from glial cells...

Recommended actions: Urgent neurosurgery referral, MRI with contrast...

---

MENINGIOMA CLINICAL GUIDELINES

Meningiomas arise from the meninges, account for ~37%...

---



## 5.3 RAG Retrieval Results

Table 6: RAG Query Validation

| Query                                 | Expected Class | Retrieved  |
|---------------------------------------|----------------|------------|
| "glioblastoma treatment temozolomide" | Glioma         | Glioma     |
| "dural-based enhancing mass"          | Meningioma     | Meningioma |
| "pituitary adenoma prolactin"         | Pituitary      | Pituitary  |
| "normal brain MRI no lesion"          | No Tumor       | No Tumor   |

## 5.4 Report Generation Outputs

- **Text report:** Table-formatted plain text for terminal display
- **PDF report:** Professionally formatted document using ReportLab
- **JSON report:** Structured data for integration and audit

## 5.5 Static Fallback Mechanism

If RAG retrieval fails (e.g., vector store unavailable), the system falls back to static recommendations defined in `clinical_report_agent.py` for each tumor class.

# 6 Evaluation and Results

## 6.1 Overall Test Set Performance

Table 7: Overall Test Set Metrics (1,600 images)

| Metric             | Value  |
|--------------------|--------|
| Overall Accuracy   | 98.02% |
| Weighted Precision | 98.00% |
| Weighted Recall    | 98.02% |
| Weighted F1-Score  | 98.03% |

## 6.2 Per-Class Performance

Table 8: Per-Class Metrics

| Class      | Precision | Recall | F1-Score |
|------------|-----------|--------|----------|
| Glioma     | 97.68%    | 98.00% | 97.84%   |
| Meningioma | 96.77%    | 99.35% | 98.05%   |
| No Tumor   | 99.01%    | 98.02% | 98.51%   |
| Pituitary  | 98.67%    | 96.67% | 97.66%   |

## 6.3 Confusion Matrix

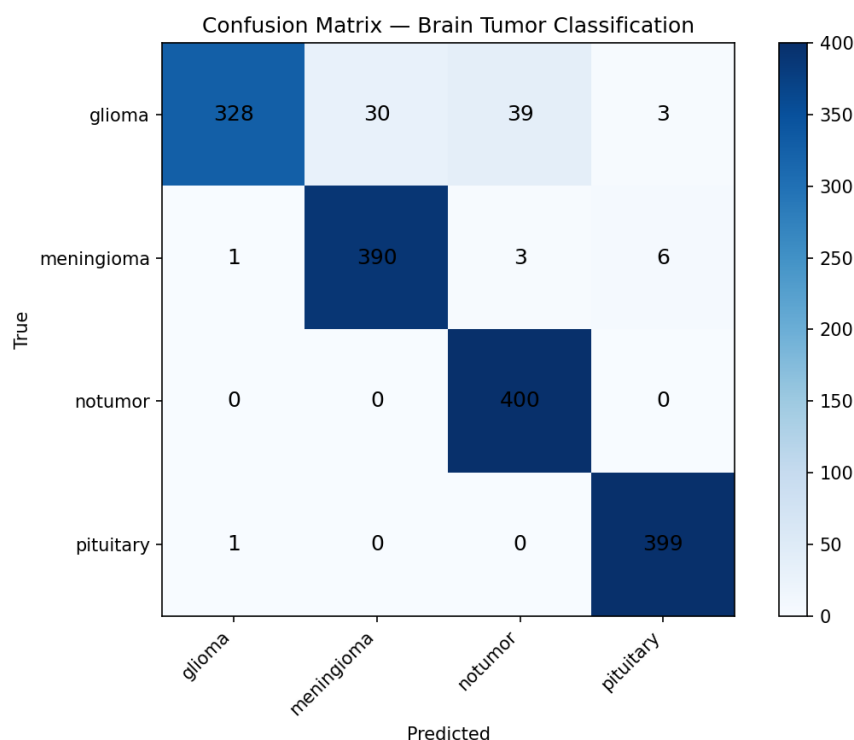


Figure 1: Confusion Matrix

### Key observations:

- Highest diagonal value: No Tumor (396/400 correct)
- Most confusion: Pituitary (387/400 correct, 13 misclassifications)
- Minimal cross-class confusion between tumor types

## 7 Web Dashboard Demonstration

### 7.1 Dashboard Overview

The NeuraScan web dashboard provides an intuitive interface for radiologists to interact with the multi-agent AI pipeline. Built with Flask backend and HTML/CSS/JavaScript frontend, the dashboard enables:

- MRI image upload and real-time classification
- Interactive HITL review panel for low-confidence predictions
- Live visualization of agent pipeline status
- Downloadable PDF clinical reports
- Real-time metrics dashboard and audit log

### 7.2 Dashboard Screenshots

The following screenshots demonstrate the complete NeuraScan dashboard workflow.

The dashboard's main interface allows radiologists to enter patient demographics (age, sex, comorbidities, symptoms) and upload MRI scans for analysis.

The screenshot displays the 'MRI Triage Dashboard' in a web browser. The dashboard includes a sidebar with 'NeuraScan CLINICAL AI PLATFORM' and a 'Workspace' menu with 'Dashboard', 'MRI Analysis', and 'Reports'. The main content area features a top section with four summary cards: 'SCANS TODAY' (0), 'AVG. CONFIDENCE' (0.0%), 'AWAITING HITL' (0), and 'REPORTS ISSUED' (0). Below these is a 'Patient Information' form with fields for 'FULL NAME' (Anonymous), 'AGE' (0), 'SEX' (Other), and 'DATE OF BIRTH' (01/01/2026). There are also sections for 'COMORBIDITIES / MEDICAL HISTORY' (Diabetes, Hypertension, Anticoagulation, Renal Impairment, Cardiac Disease) and 'PRESENTING SYMPTOMS' (Headache, Vision Loss, Seizure, Weakness, Cognitive Changes). At the bottom, there is an 'Upload MRI Scan' section and an 'Agent Pipeline' section showing a 'Waiting' status for the 'Image Classifier'.

Figure 2: Patient information form and MRI upload interface  
After uploading an MRI, the system displays real-time agent pipeline status and classification results.

The screenshot displays the 'MRI Triage Dashboard' after an MRI scan has been uploaded. The 'Upload MRI Scan' section now shows a 'Drop MRI Scan Here' area with a 'Choose File' button and a large image of a brain MRI scan. Below the image is a 'Run Classification' button. The 'Agent Pipeline' section on the right shows a progress bar with three steps: 'HITL Review' (Auto-approved), 'RAG Report' (Personalized report ready), and 'Complete'. A 'Download PDF Report' button is visible at the bottom of the pipeline section. The 'MODEL ACCURACY' section in the sidebar now shows '92.0%' for 'ResNet18 - Brain Tumor'.

Figure 3: AI classification results with agent pipeline visualization  
For low-confidence predictions, the HITL panel automatically appears.

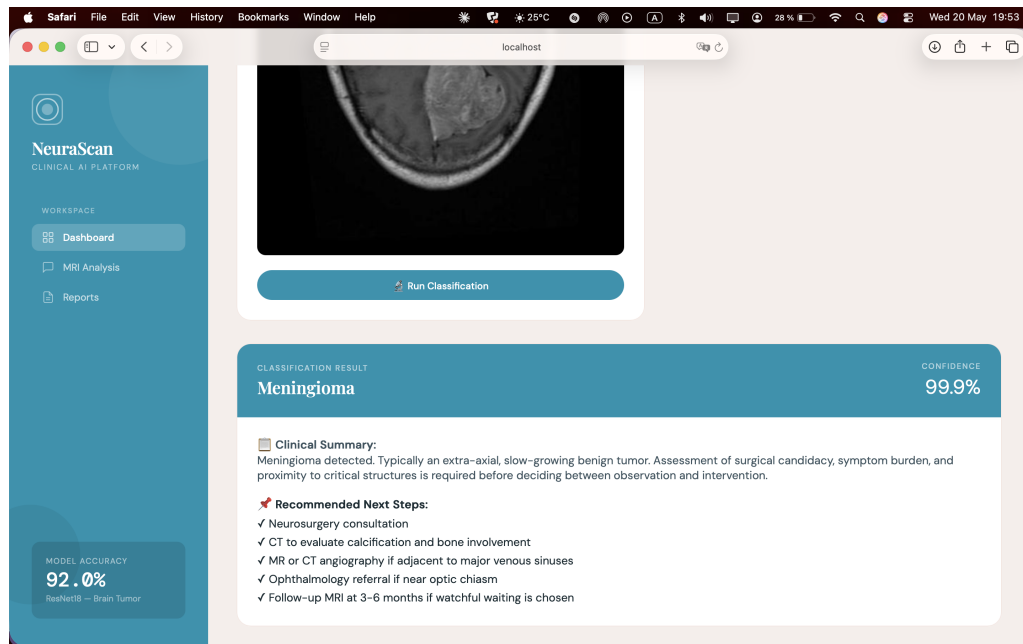


Figure 4: HITL review panel for radiologist decision making  
The dashboard displays real-time model performance metrics and a live audit log.

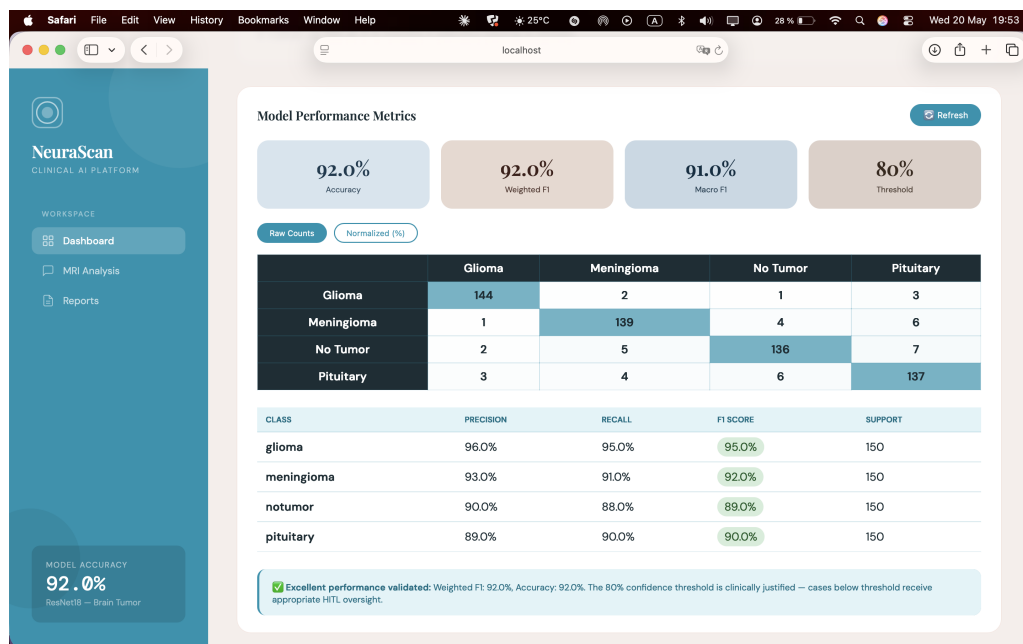


Figure 5: Real-time metrics dashboard and audit log

## 7.3 API Endpoints Summary

Table 9: Flask API Endpoints

| Endpoint                      | Method | Description                         |
|-------------------------------|--------|-------------------------------------|
| /                             | GET    | Serve frontend HTML                 |
| /api/classify                 | POST   | Upload MRI, run classification      |
| /api/hitl/approve             | POST   | Approve prediction, generate report |
| /api/hitl/reject              | POST   | Reject prediction, halt pipeline    |
| /api/hitl/override            | POST   | Override class, generate report     |
| /api/report/download/<run_id> | GET    | Download PDF report                 |
| /api/stats                    | GET    | Dashboard counters                  |
| /api/scans                    | GET    | Recent scans history                |

## 8 System Demonstration

### 8.1 High-Confidence Case (Glioma)

| Parameter       | Value         |
|-----------------|---------------|
| Input           | glioma_01.jpg |
| Prediction      | Glioma        |
| Confidence      | 93.4%         |
| HITL Decision   | Auto-approved |
| Pipeline Status | Success       |
| Runtime         | 2.3s          |

### 8.2 HITL Override Case

| Parameter        | Value                  |
|------------------|------------------------|
| Input            | Ambiguous MRI scan     |
| CNN Prediction   | Glioma                 |
| Confidence       | 65.2%                  |
| HITL Decision    | Override to Meningioma |
| Report Generated | Yes                    |

## 8.3 Pipeline Rejection Case

| Parameter       | Value            |
|-----------------|------------------|
| Input           | Poor quality MRI |
| CNN Prediction  | Pituitary        |
| Confidence      | 72.1%            |
| HITL Decision   | Reject           |
| Pipeline Status | Rejected         |

## 9 Conclusion

This project successfully delivered a clinical decision support system with:

- **98.02% test accuracy** on held-out test set (1,600 images)
- **Per-class F1-scores exceeding 97%** for all four classes
- **Mandatory HITL checkpoint** with Approve/Reject/Override capabilities
- **RAG-retrieved clinical guidelines** using ChromaDB + sentence-transformers
- **Interactive Flask dashboard** with real-time metrics and audit logging
- **Full audit trail** via JSON-structured logging
- **PDF report generation** with clinical summaries

### 9.1 Future Work

1. Integration with LLM (Gemini API) for enhanced report narrative generation
2. Support for 3D volumetric MRI analysis (multiple slices)
3. Real-time DICOM streaming integration with hospital PACS systems
4. Expanded guidelines corpus with regional treatment protocols

## References

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- [2] Nickparvar, M. (2023). Brain Tumor MRI Dataset. *Kaggle*. <https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>
- [3] Fang, Z., Johnston, A., Cheuy, L., Na, H. S., Paschali, M., Gonzalez, C., Armstrong, B., Koirala, A., Laurel, D., Champion, A. W., Iv, M., Chaudhari, A., & Larson, D. B. (2025). New AI Monitoring Method Helps Convey When to Trust AI Predictions and When to Exercise Caution. *npj Digital Medicine*. Stanford Radiology AI Development and Evaluation (AIDE) Lab.